

N° ECUE	XA4S513								
Title	Hydrodynamic in reactors								
ECTS		Lecture(h) CM		Tutorials (h) TD		Practical works (h) TP		Project (h)	
Description									
This course aims to introduce the notions of mixing efficiency, macro and micro mixing, into a reactor (or exchanger) and the consequences on the performance and optimal design of units. The teaching has two main parts: - Theory of the reactors: residence time distribution, behavior of ideal reactors (plug-flow and Continuous Ideally Stirred-Tank Reactor (CISTR)), flow anomalies, behavior of real reactors and simplified models of flow (jCISTR in series...) - Mixing notions to illustrate an upscaling problem									
Key Words	Mixing, reactor, hydrodynamic								
Type of Evaluation	Exam Practical work								